



Nonlinear Programming

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EgyE 332

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Outline

1. Motivational Example
2. Representative Algorithms
    a. Newton Methods (Unconstrainted NLP)
    b. Evolutionary Algorithms (Unconstrained NLP)
    c. Lagrange Multipliers (Constrained NLP)
    d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
    a. Cascaded Dams Scheduling
    b. Wind Farm Layout Optimization

Nonlinear Programming

Solar PV Max Power Output

https://github.com/kspilario/EgyE332_opti/blob/main/NLP%20Problems/solar_max_power.ipynb

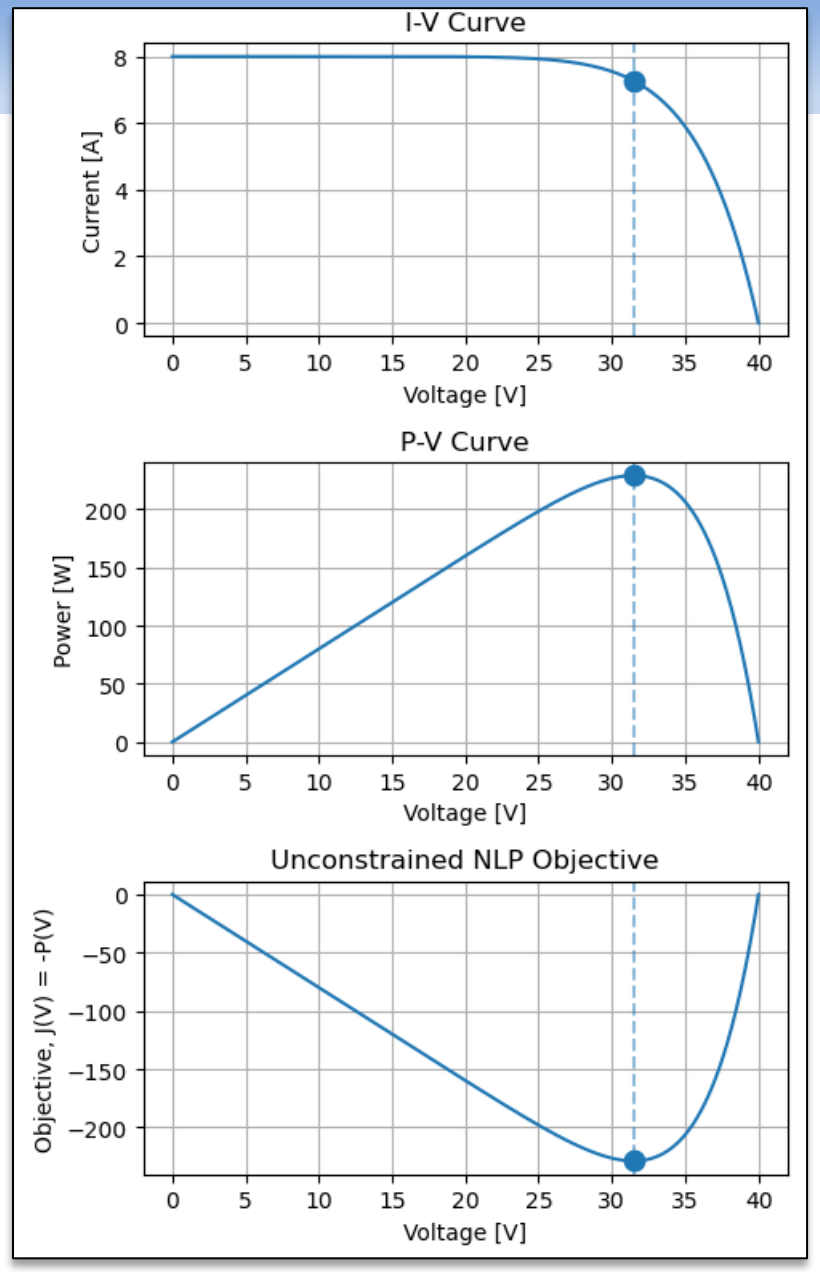
The electrical output of a solar PV depends on the voltage at which it is operated.

For a given irradiance and temperature, a PV module has an I–V characteristic, $I = f(V)$. A simple approximation is given by:

$$I(V) = I_{SC}\left[1 - \left(\frac{V}{V_{OC}}\right)^{a}\right]$$

Where I_{SC} is the short-circuit current, V_{OC} is the open-circuit voltage, and a is a shape parameter.

Since $P = IV$, we need to solve for the V that gives the maximum power P given a fixed a, I_{SC} *and* V_{OC}.



Mathematical Programs

In general, the method for solving mathematical models is highly dependent on the **type of model**:

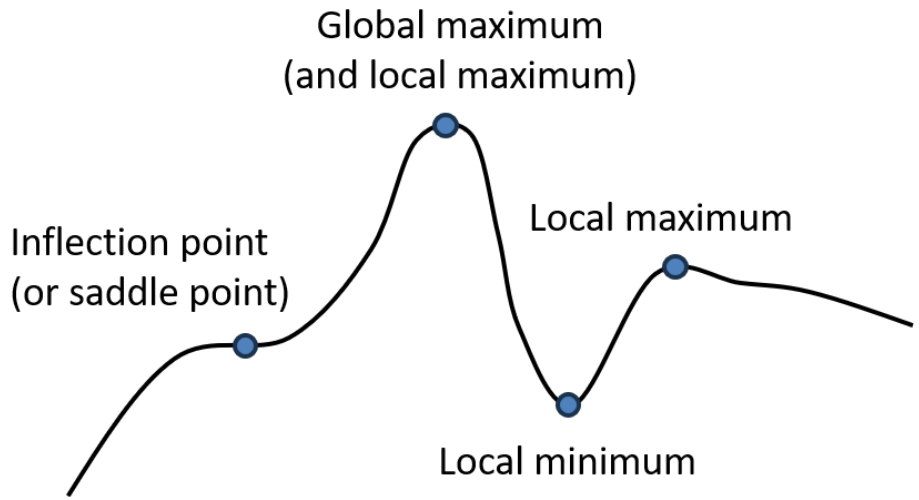
TYPE	Objective Function	Constraints	Decision Variables
Linear Program (**LP**)	Linear	Linear	Continuous
Nonlinear Program (**NLP**)	At least the objective function or 1 constraint is nonlinear		Continuous
Integer Program (**IP**)	Linear or nonlinear	Linear or nonlinear	All Integer
Mixed-integer Linear Program (**MILP**)	Linear	Linear	Some Integer; Others Continuous
Mixed-integer Nonlinear Program (**MINLP**)	At least the objective function or 1 constraint is nonlinear		Some Integer; Others Continuous

We'll discuss
BFGS (Unconstrained)
GA & PSO (Unconstrained) and
IPOPT (Constrained)

Source: Chap.14; El-Halwagi, M., 2012. *Sustainable Design Through Process Integration.* USA: Elsevier Inc.

*The word "program" is due to George Dantzig (1974).

Unconstrained NLP



Local minimum

A point $\boldsymbol{x}^*$ such that no other point in the *vicinity* of $\boldsymbol{x}^*$ yields a value of $f(\boldsymbol{x})$ less than $f(\boldsymbol{x}^*)$:

$$f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*)$$

Global minimum

A point $\boldsymbol{x}^*$ such that $f(\boldsymbol{x}) \geq f(\boldsymbol{x}^*)$ holds for any $\boldsymbol{x}$ in the n-dimensional space of $\boldsymbol{x}$.

Conditions for Optimality in Unconstrained NLPs

A point $\boldsymbol{x}^*$ is an **optimal solution** to an unconstrained NLP:

- ONLY IF $f(\boldsymbol{x})$ is twice differentiable at $\boldsymbol{x}^*$. **NECESSARY**
- ONLY IF $\boldsymbol{\nabla} f(\boldsymbol{x}^*) = 0$ or $\boldsymbol{x}^*$ is a stationary point. **NECESSARY**
- IF $\boldsymbol{H}(\boldsymbol{x}^*)$ is *positive-definite* for a minimum to exist, and *negative-definite* for a maximum to exist. **SUFFICIENT**

$\boldsymbol{H}(\boldsymbol{x})$ is the Hessian matrix.

$$\mathbf{H}_f = \begin{bmatrix} \frac{\partial^2 f}{\partial x_1^2} & \frac{\partial^2 f}{\partial x_1 \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_1 \partial x_n} \\ \frac{\partial^2 f}{\partial x_2 \partial x_1} & \frac{\partial^2 f}{\partial x_2^2} & \cdots & \frac{\partial^2 f}{\partial x_2 \partial x_n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f}{\partial x_n \partial x_1} & \frac{\partial^2 f}{\partial x_n \partial x_2} & \cdots & \frac{\partial^2 f}{\partial x_n^2} \end{bmatrix}$$

$\boldsymbol{H}(\boldsymbol{x})$ is…	If all the eigenvalues of $\boldsymbol{H}(\boldsymbol{x})$ are…
Positive-definite	> 0
Positive-semidefinite	≥ 0
Negative-definite	< 0
Negative-semidefinite	≤ 0

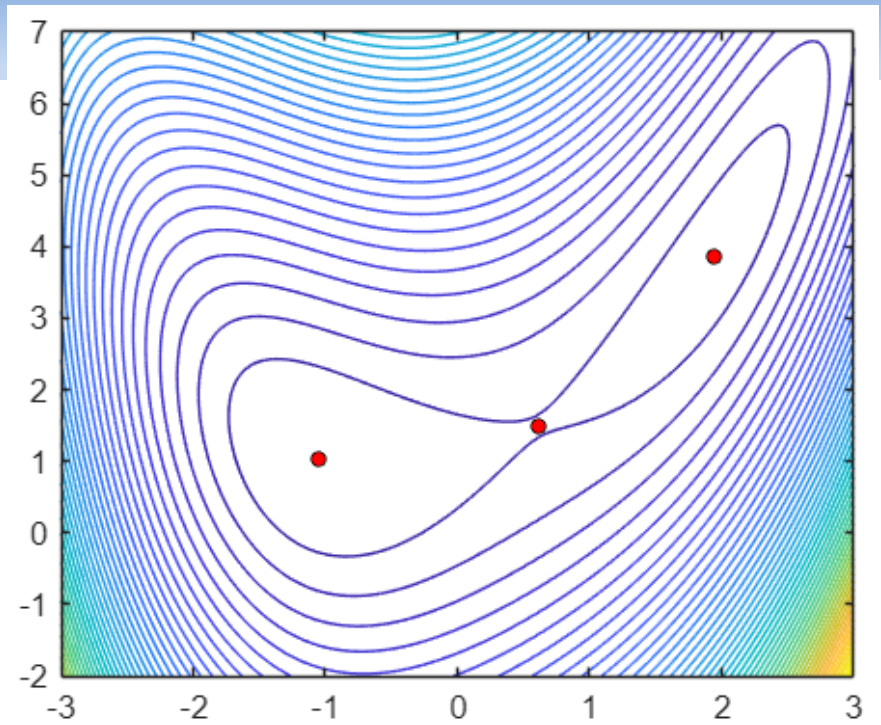
Unconstrained NLP

Example:

$$f(x_1, x_2) = 4 + 4.5x_1 - 4x_2 + x_1^2 + 2x_2^2 - 2x_1x_2 + x_1^4 - 2x_1^2x_2$$

This function has three stationary points $(1.941, 3.854)$, $(-1.053, 1.028)$, and $(0.6117, 1.4929)$. Evaluate the optimality at these points.

$$\boldsymbol{H}(x_1, x_2) = \begin{bmatrix} 2 + 12x_1^2 - 4x_2 & -2 - 4x_1 \\ -2 - 4x_1 & 4 \end{bmatrix}$$



Stationary point	$f(\boldsymbol{x})$	$f'(\boldsymbol{x})$	$\boldsymbol{H}(\boldsymbol{x})$	Eigenvalues		Evaluation
$(1.941, 3.854)$	0.9856	0.00	$\begin{bmatrix} 31.79 & -9.76 \\ -9.76 & 4.00 \end{bmatrix}$	0.9128	34.8810	**Local minimum**
$(-1.053, 1.028)$	−0.5134	0.00	$\begin{bmatrix} 11.19 & 2.21 \\ 2.21 & 4.00 \end{bmatrix}$	3.3743	11.8194	**Local minimum**
$(0.6117, 1.4929)$	2.8091	0.00	$\begin{bmatrix} 0.52 & -4.45 \\ -4.45 & 4.00 \end{bmatrix}$	-2.5161	7.0346	**Saddle point**

Unconstrained NLP

Unlike an LP, solving an NLP is hard because a minimum is **not guaranteed** to be the global minimum.

But if the NLP is **convex**, we have the following result.

Theorem

For an NLP whose objective function $f(\boldsymbol{x})$ is a **convex function** and each inequality constraint $g(\boldsymbol{x})$ is a convex function so that they form a **convex set**:

The local minimum of $f(\boldsymbol{x})$ is also a global minimum.

Analogously, if $f(\boldsymbol{x})$ is a **concave function** and the constraints $g(\boldsymbol{x})$ still form a **convex set**, then:

The local maximum of $f(\boldsymbol{x})$ is also a global maximum.

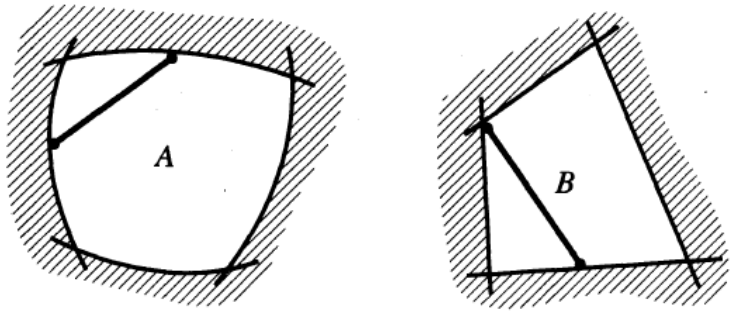
Definition: Convex Set

For every pair of points $\boldsymbol{x}_1$ and $\boldsymbol{x}_2$ in a convex set, the point $\boldsymbol{x}$ given by a linear combination of the two points:

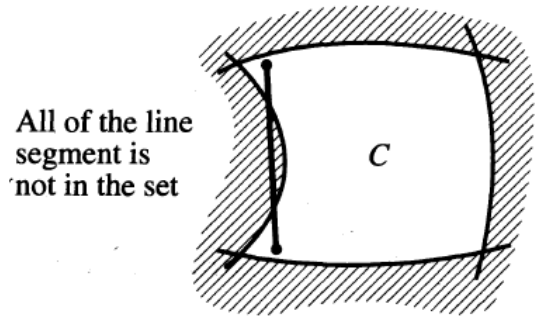
$$\boldsymbol{x} = \gamma \boldsymbol{x}_1 + (1-\gamma)\boldsymbol{x}_2, \qquad 0 \le \gamma \le 1$$

must also be in the set. Also, the intersection of any number of convex sets is a convex set.

Examples of a Convex Set



Example of a Nonconvex Set



Unconstrained NLP

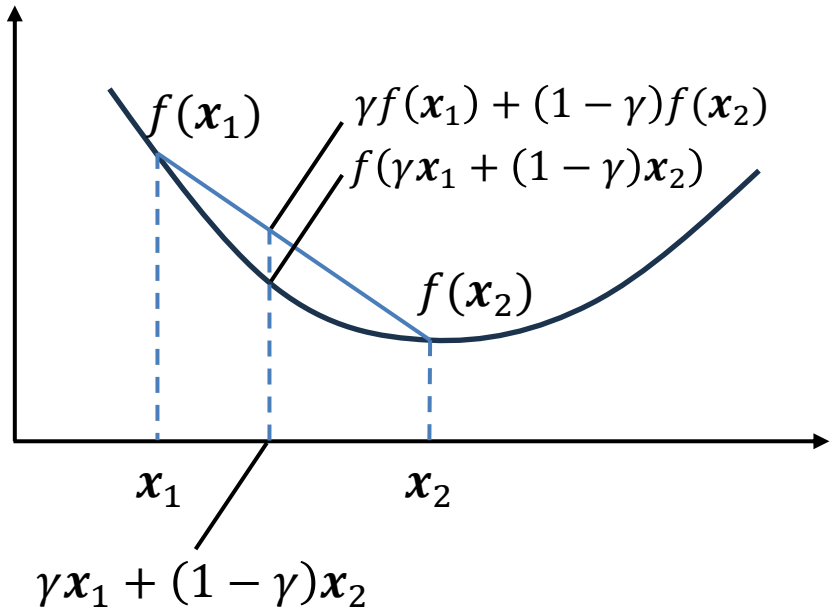
Definition: Convex Function

A function $f(\boldsymbol{x})$ defined on a convex set F is said to be a **convex function** if the following relation holds:

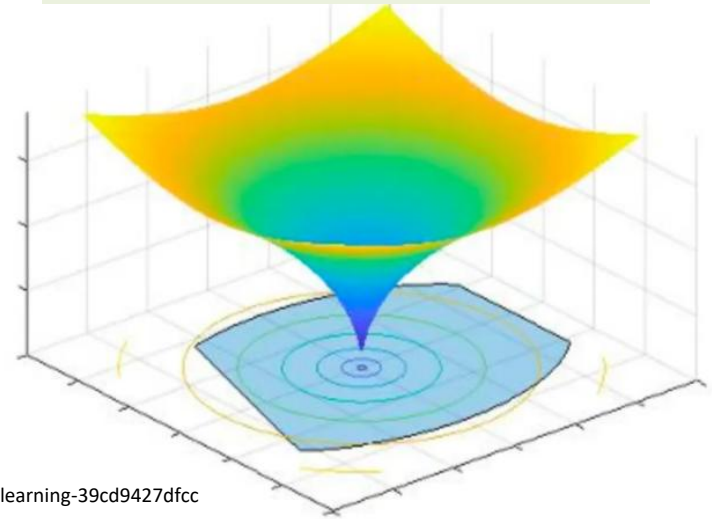
$$f(\gamma \boldsymbol{x}_1 + (1-\gamma)\boldsymbol{x}_2) \leq \gamma f(\boldsymbol{x}_1) + (1-\gamma) f(\boldsymbol{x}_2)$$

Where $0 \leq \gamma \leq 1$.

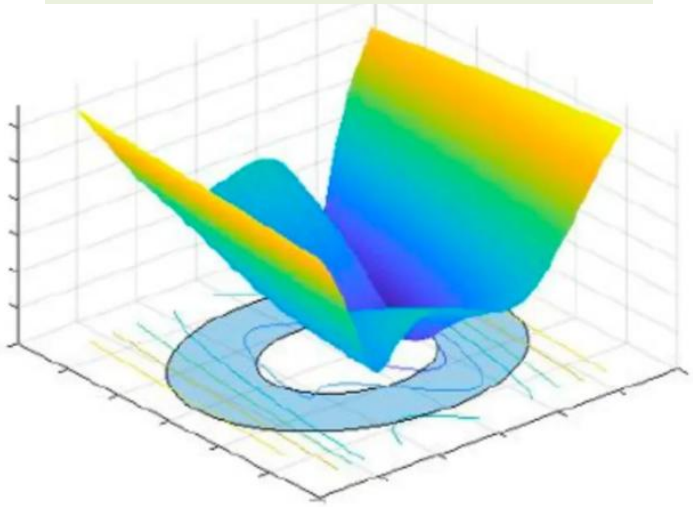
- If $f(\boldsymbol{x})$ is convex, then $-f(\boldsymbol{x})$ is concave.
- If only the inequality sign holds ($<$), the function is said to be *strictly convex*.
- If $f(\boldsymbol{x})$ is strictly convex, then $-f(\boldsymbol{x})$ is strictly concave.



A **convex** objective with **convex** constraints.



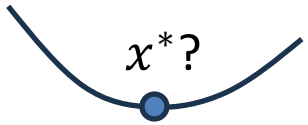
A **nonconvex** objective with **nonconvex** constraints.



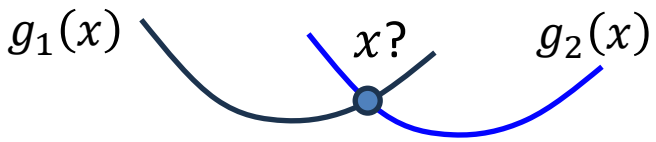
Source: https://rumn.medium.com/convex-vs-non-convex-functions-why-it-matters-in-optimization-for-machine-learning-39cd9427dfcc

Unconstrained NLP: Newton's Methods

Minimize: $f(x)$



Recall: Newton's Method for Root-Finding



Given: $f(x) = g_1(x) - g_2(x) = 0$

Initialization: One initial guess (x_0) and a Tolerance, Tol

1: Set $x_{old} \leftarrow \text{Inf.}$

2: Set $x_{new} \leftarrow x_0$.

3: **WHILE** $|x_{new} - x_{old}| > Tol$:

4: Assign: $x_{old} \leftarrow x_{new}$.

5: Assign: $x_{new} \leftarrow x_{old} - \frac{f(x_{old})}{f'(x_{old})}$.

6: **END WHILE**

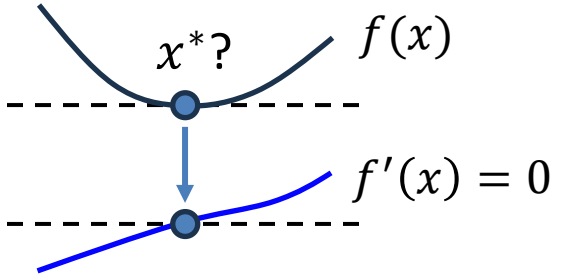
7: **OUTPUT** x_{new}

$$x_{k+1} = x_k - \frac{f(x_k)}{f'(x_k)}$$

Newton's Method for Optimization

Minimize: $f(x)$

Find the roots: $f'(x) = 0$



Newton's Method

$$x_{k+1} = x_k - \frac{f'(x_k)}{f''(x_k)}$$

$$f''(x_k) \approx \frac{f'(x_{k-1}) - f'(x_k)}{x_{k-1} - x_k}$$

2nd Order finite difference

Quasi-Newton Method

$$x_{k+1} = x_k - \frac{(x_{k-1} - x_k)f'(x_k)}{f'(x_{k-1}) - f'(x_k)}$$

Unconstrained NLP: Newton's Methods

This formula only works for one-dimensional search spaces:

Quasi-Newton Method

$$x_{k+1} = x_k - \frac{(x_{k-1} - x_k)f'(x_k)}{f'(x_{k-1}) - f'(x_k)}$$

$$f'(x_k) \Rightarrow \nabla f(\boldsymbol{x}_k)$$

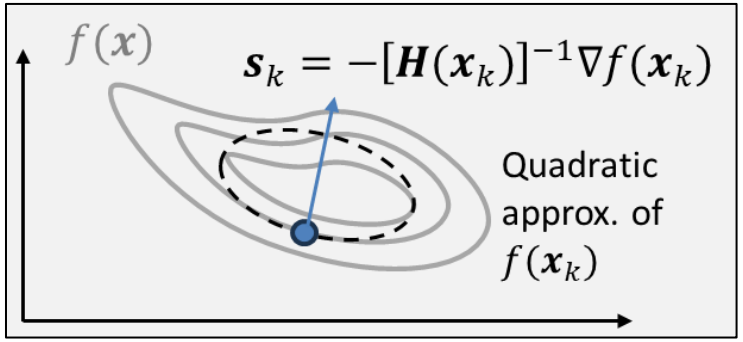
$$f''(x_k) \Rightarrow \boldsymbol{H}(\boldsymbol{x}_k)$$

L-BFGS Algorithm

(Limited-memory Broyden-Fletcher-Goldfarb-Shanno)

$$\boldsymbol{x}_{k+1} = \boldsymbol{x}_k - \alpha_k[\boldsymbol{H}(\boldsymbol{x}_k)]^{-1}\nabla f(\boldsymbol{x}_k)$$

α_k = step size at the kth iteration
$\boldsymbol{s}_k$ = search direction
$\nabla f(\boldsymbol{x}_k)$ = direction of steepest descent
$\boldsymbol{H}(x_k)$ = Hessian matrix



Steps:

- Make an initial guess, $\boldsymbol{x}_0$, and initial guess of inverse Hessian $\boldsymbol{\Gamma}_0 = \boldsymbol{I}$. [*steepest descent*]
- Calculate *search direction*, $\boldsymbol{s}_k = -\boldsymbol{\Gamma}_k \nabla Cost(\boldsymbol{x}_k)$
- Calculate step size, α, using line search.
- Update $\boldsymbol{x}_{k+1} = \boldsymbol{x}_k + \alpha \boldsymbol{s}_k$.
- Update $\boldsymbol{\Gamma}_k$ to $\boldsymbol{\Gamma}_{k+1}$.

In addition, BFGS only *approximates* the Hessian and computes a *step size* using line search.

$\boldsymbol{\Gamma}_k$ = kth approximation of the inverse Hessian

$$\boldsymbol{\Gamma}_{k+1} = \left(\boldsymbol{I} - \frac{\boldsymbol{s}_k \boldsymbol{y}_k^T}{\boldsymbol{y}_k^T \boldsymbol{s}_k}\right)\boldsymbol{\Gamma}_k\left(\boldsymbol{I} - \frac{\boldsymbol{y}_k \boldsymbol{s}_k^T}{\boldsymbol{y}_k^T \boldsymbol{s}_k}\right) + \frac{\boldsymbol{s}_k \boldsymbol{s}_k^T}{\boldsymbol{y}_k^T \boldsymbol{s}_k}$$

where $\boldsymbol{y}_k = \nabla f(\boldsymbol{x}_{k+1}) - \nabla f(\boldsymbol{x}_k)$

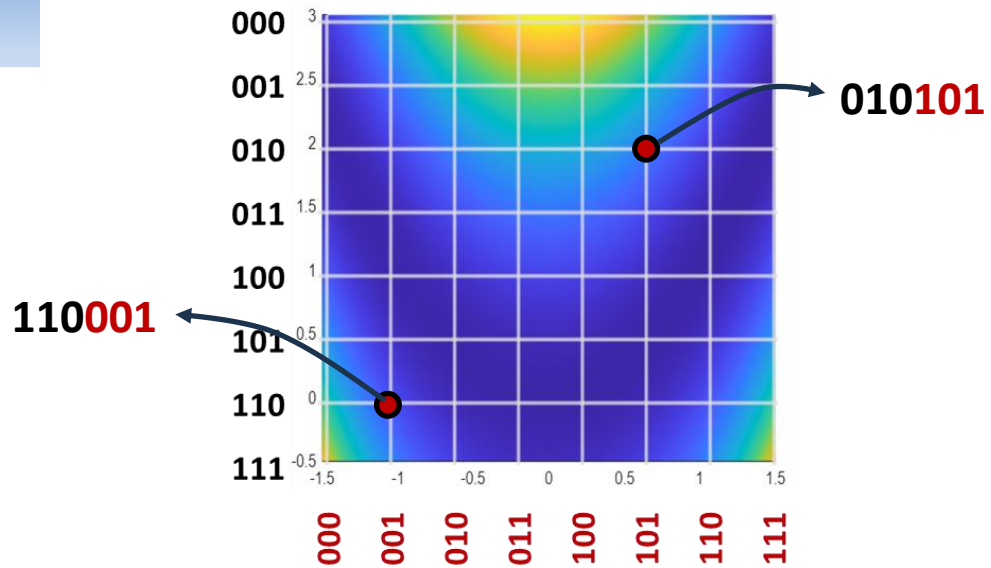
Outline

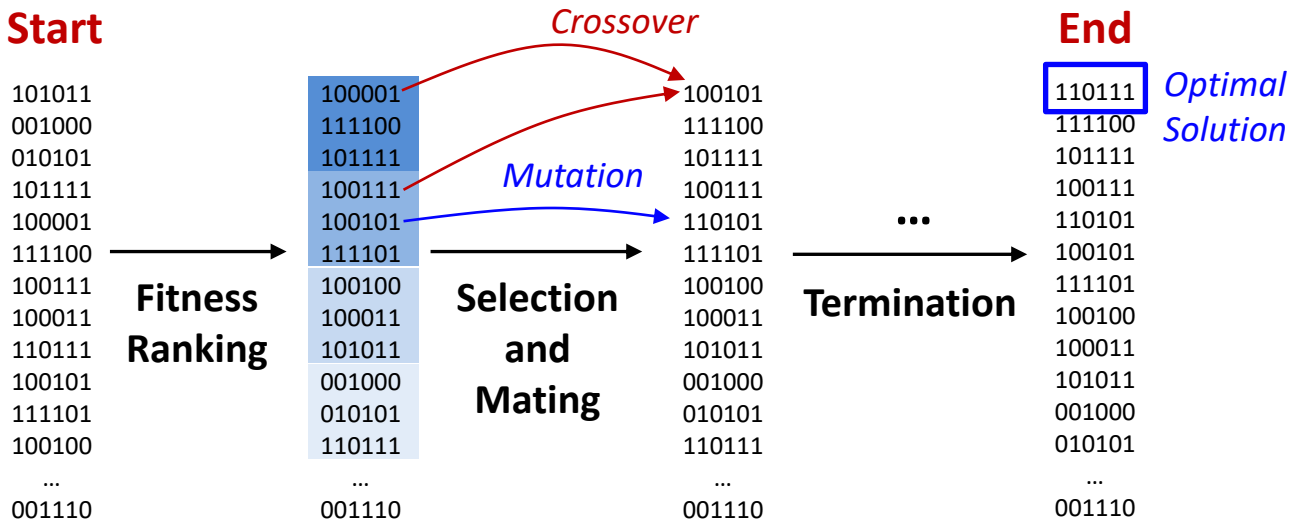
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2. Representative Algorithms
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Evolutionary Algorithms

Genetic Algorithm

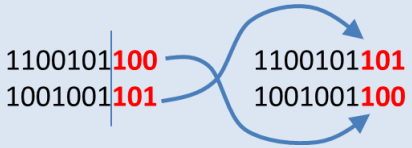
- John Holland, *Adaptation in Natural and Artificial Systems* (1975)
- Inspired by organisms “optimized” through years of evolution via **survival of the fittest**.
- Represent solutions as **chromosomes** (e.g. bit strings).
- In every generation, chromosomes are allowed to mate (**crossover**) and mutate (**mutation**) until convergence.





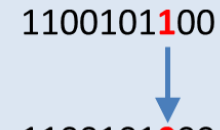
Crossover

- Promotes **exploitation.**
- Breed parents.
- The higher the fitness, the higher chance of being selected as parent.



Mutation

- Promotes **exploration.**
- Select a chromosome, then randomly change 1 or some bits.



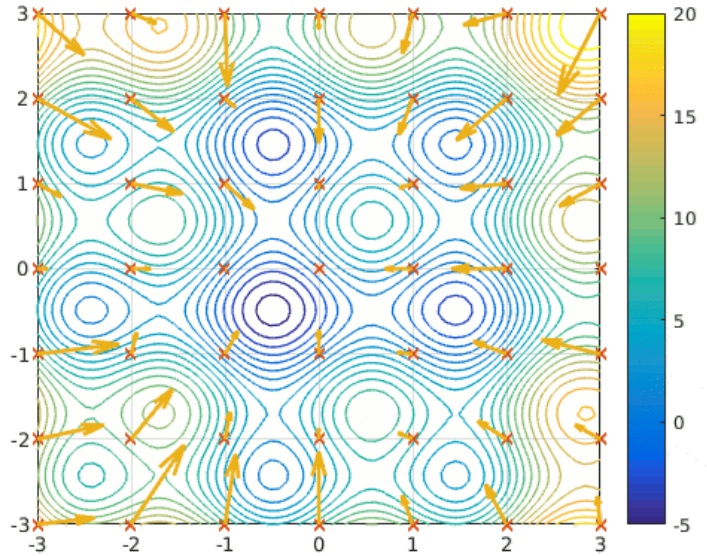
Evolutionary Algorithms

Particle Swarm Optimization

- James Kennedy, Russell Eberhart, Yuhui Shi (1995)
- Inspired by **swarming of bees** and **flocks of birds** that can navigate together by sharing nearby information.
- Update solutions by computing their velocities, $\mathbf{V}_\mathrm{p}^i$:

$$\mathbf{x}_\mathrm{p}^i = \mathbf{x}_\mathrm{p}^{i-1} + \mathbf{V}_\mathrm{p}^i$$

$$\mathbf{V}_\mathrm{p}^i = \mathrm{c}_0\mathbf{V}_\mathrm{p}^{i-1} + \mathrm{c}_1 \times \mathrm{rand} \times \left(\mathbf{x}_p^{\mathrm{best}} - \mathbf{x}_\mathrm{p}^{i-1}\right) + \mathrm{c}_2 \times \mathrm{rand} \times \left(\mathbf{x}_\mathrm{g}^{\mathrm{best}} - \mathbf{x}_\mathrm{p}^{i-1}\right)$$

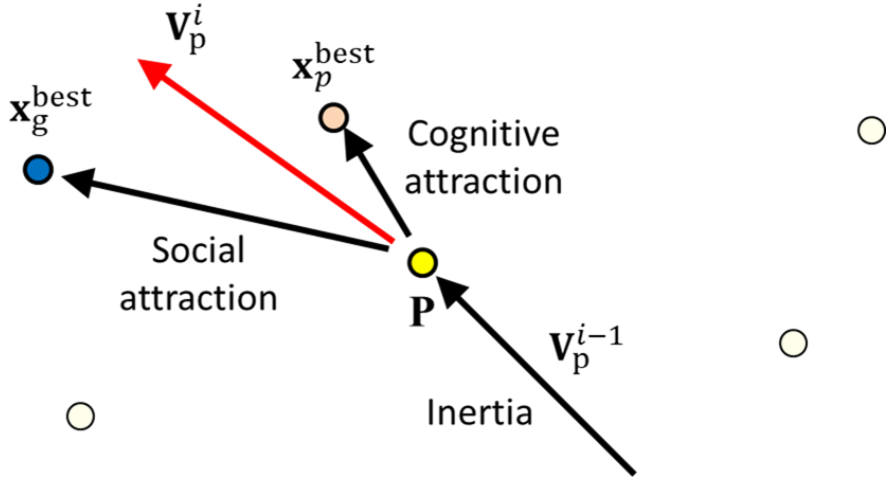


c_0 **Inertia weight**	$\mathbf{V}_\mathrm{p}^i$ Velocity for the current iteration (i)
c_1 **Cognitive parameter**	$\mathbf{V}_\mathrm{p}^{i-1}$ Velocity from the previous iteration $(i-1)$
c_2 **Social parameter**	$\mathbf{x}_\mathrm{p}^{i-1}$ Previous position of the particle $(i-1)$

$\mathbf{x}_p^{\mathrm{best}}$ Best position the individual particle p encountered so far

$\mathbf{x}_\mathrm{g}^{\mathrm{best}}$ Global best position encountered by whole swarm so far

rand a random real number from 0 to 1 that's different every time.

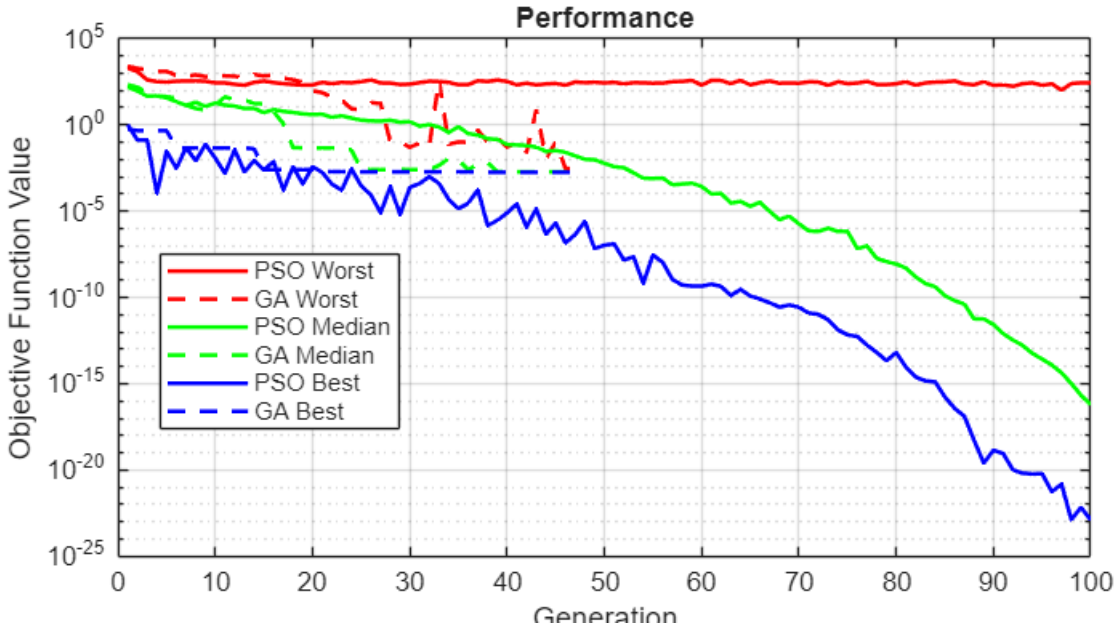


Evolutionary Algorithms

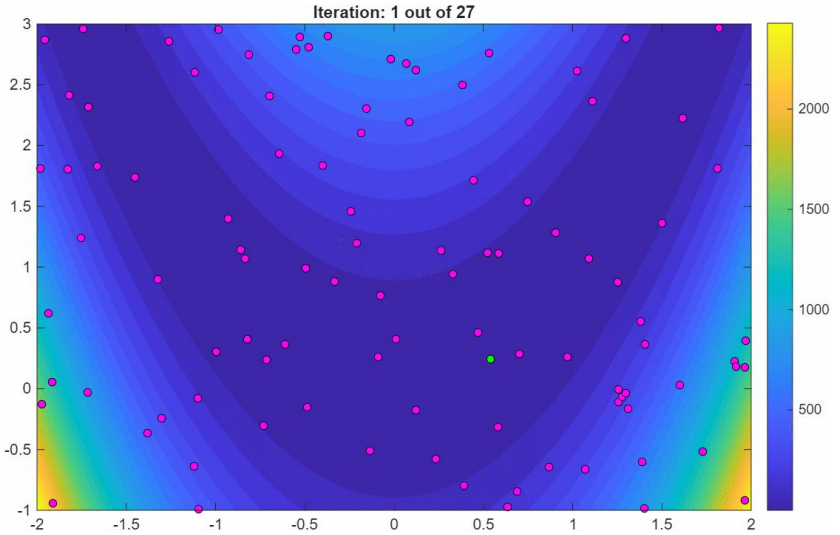
Example:

Find the minimum of the **Rosenbrock function** using GA and PSO with the following settings:

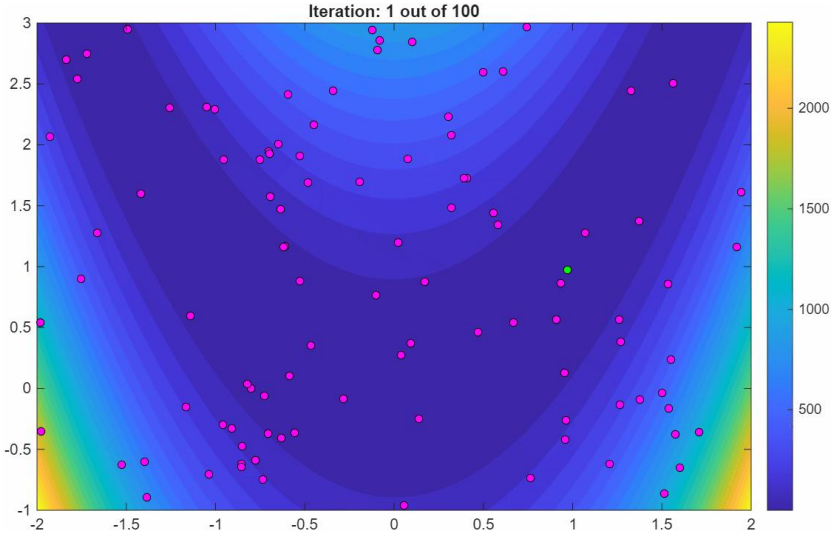
- Population size: 100 (both)
- No. of iterations: 100 (both)
- GA: Crossover rate = 1.0, Mutation rate = 0.01
- PSO: Cognitive = 2.0, Social = 2.0, inertia = 0.9 to 0.1



Genetic Algorithm



Particle Swarm Optimization



Evolutionary Algorithms

By August 2020, there are already more than 200 different nature-inspired optimization algorithms!

Source: Tzanetos et al. (Aug 2020). A comprehensive database of Nature-Inspired Algorithms. Data in Brief. Elsevier. doi: 10.1016/j.dib.2020.105792

Ant Colony Optimization
Bacterial Foraging Optimization
BeeHive
Bee Colony Optimization
The Bees Algorithm
Artificial Bee Colony
Bee Swarm Optimization
Hunting Search
Bat Algorithm
Cockroach Swarm Optimization
Eurygaster Algorithm
Blind naked Mole-rat Algorithm
OptBees
Wolf Search
African Wild Dog Algorithm
Seven-spot Ladybird Optimization
Penguins Search Optimization Algorithm
Grey Wolf Optimizer
Chicken Swarm Optimization
Cockroach Swarm Evolution
Butterfly Optimization Algorithm
Whale Optimization Algorithm
Bird Swarm Algorithm
Dolphin Swarm Optimization Algorithm
Raven Roosting Optimisation Algorithm
Spotted Hyena Optimizer
Whale Swarm Algorithm
Meerkat Clan Algorithm
Yellow Saddle Goatfish
Sailfish Optimizer
Manta Ray Foraging Optimization
Tunicate Swarm Algorithm
Fish-swarm Algorithm
Glowworm Swarm-based Optimization
Roach Infestation Optimization
Bumblebees
Cuckoo Search
Dolphin Partner Optimization
Firefly Algorithm
Locust Swarms

Termite Colony Optimization
Krill Herd
Social Spider Optimization
Swallow Swarm Optimization Algorithm
Spider Monkey Optimization Algorithm
Wolf Pack Algorithm
Earthworm Optimization Algorithm
African Buffalo Optimization
Elephant Herding Optimization
Elephant Search Algorithm
See-See Partridge Chicks Optimization
Dolphin Swarm Algorithm
Lion Optimization Algorithm
Emperor Penguin Optimizer
Coyote Optimization Algorithm
Anglerfish Algorithm
Dominion Algorithm
Emperor Penguins Colony
Blue Monkey
Black Widow Optimization
Black Widow Optimization Algorithm
Stochastic Diffusion Search
Particle Swarm Optimization
Sheep Flocks Heredity Model Algorithm
Bacterial Swarming Algorithm
Jumping Frogs Optimization
Fish School Search
Artificial Searching Swarm Algorithm
Hierarchical Swarm Model
Bumble Bees Mating Optimization
Migrating Birds Optimization
Weightless Swarm Algorithm
Pigeon-inspired Optimization
Animal Migration Optimization
Moth Flame Optimization
Cockroach Colony Optimization
Monarch Butterfly Optimization
Dragonfly Algorithm
Grasshopper Optimization Algorithm
Salp Swarm Algorithm

Selfish Herd Optimizer
Bison
Biology Migration Algorithm
Lion Pride Optimization Algorithm
Wildebeests Herd Optimization
Artificial Social Cockroaches
Meerkat-inspired Algorithm
Seagull Optimization Algorithm
Vocalization of Humpback Whale Optimization Algorithm
Military Dog Optimizer
Fitness Dependent Optimizer
Collective Animal Behavior
Side-Blotched Lizard Algorithm
Marriage in Honey Bees
Bacterial Chemotaxis
Honey-bees Mating Optimization
Shuffle frog-leaping Algorithm
Cat Swarm Optimization
Group Search Optimizer
Monkey Search
Japanese tree frogs
Biogeography-based Optimization
Bee Collecting Pollen Algorithm
Viral Systems
Monkey Algorithm
Fruit Fly Optimization
Mosquitos Oviposition
Bat Sonar Algorithm
Hoopoe Heuristic Optimization
Lion Pride Optimizer
The Great Salmon Run
The Lion's Algorithm
Superbug Algorithm
Keshtel Algorithm
Swine Influenza Models Based Optimization
Dolphin Echolocation
Green Herons Optimization Algorithm
Egyptian Vulture Optimization Algorithm
Jumper Firefly Algorithm
Cuttlefish Algorithm

Mussels Wandering Optimization
Prey-Predator Algorithm
Magnetotactic Bacteria Optimization Algorithm
Symbiotic Organisms Search
Coral Reefs Optimization Algorithm
Bird Mating Optimizer
Competition over Resources
Ant Lion Optimizer
Artificial Algae Algorithm
Jaguar Algorithm
Sperm Whale Algorithm
Virus Colony Search
Shark Smell Optimization
Crow Search Algorithm
Camel Algorithm
Virulence Optimization Algorithm
Mosquito flying optimization
Red Deer Algorithm
Rhino Herd
Polar Bear Optimization Algorithm
Fish Electrolocation Optimization
Satin Bowerbird Optimizer
Artificial Butterfly Optimization
Beetle Antennae Search Algorithm
Killer Whale Algorithm
Laying Chicken Algorithm
Cyclical Parthenogenesis Algorithm
Cyclical Parthenogenesis Algorithm
Beetle Swarm Optimization Algorithm
Eagle Piercing Optimizer
Rhinoceros Search Algorithm
Dolphin Pod Optimization
Owl Search Algorithm
Pontogammarus Maeoticus Swarm Optimization
Barnacles Mating Optimizer
Mouth Brooding Fish algorithm
Pity Beetle Algorithm
Cheetah Based Optimization Algorithm
Circular Structures of Puffer Fish
Water-Tank Fish Algorithm
Neuronal Communication

Cheetah Chase Algorithm
Falcon Optimization Algorithm
Flying Squirrel Optimizer
Harris Hawks Optimization
Sooty Tern Optimization Algorithm
Donkey and Smuggler Optimization Algorithm
Artificial Feeding Birds
Squirrel Search Algorithm
Andean Condor Algorithm
Genetic Algorithm
Bald Eagle Search
Raccoon Optimization Algorithm
Sea Lion Optimization Algorithm
Electric Fish Optimization
Urban Pigeon-Inspired Optimiser
Xerus Optimization Algorithm
Artificial Ecosystem-based Optimization
Marine Predators Algorithm
Shuffled Shepherd Optimization Algorithm
Water Strider Algorithm
Photosynthetic Algorithm
Saplings Growing up Algorithm
Invasive Weed Optimization
Plant Growth Optimization
Paddy Field Algorithm
Plant Propagation Algorithm
Artificial Plant Optimization Algorithm
Ecology-inspired Optimization Algorithm
Flower Pollination Algorithm
Physarum Optimization
Root Mass Optimization
Forest Optimization Algorithm
Root Growth Algorithm
Seed-based Plant Propagation Algorithm
Tree Seed Algorithm
Runner-root Algorithm
Natural Forest Regeneration
Waterweeds Algorithm
Root Tree Optimization Algorithm
Artificial Tree Algorithm
Mushroom Reproduction Optimization
Invasive Tumor Growth Optimization
Allostatic Optimization
Artificial Infectious Disease Optimization

Chemotherapy Science Algorithm
Kidney-inspired Algorithm
Artificial Coronary Circulation System
Artificial Ecosystem-based Optimization
Space Gravitational Algorithm
Big Bang - Big Crunch
Gravitational Search Algorithm
Galaxy-based Search Algorithm
Gravitational Interactions Optimization
Black Hole
Space Gravity Optimization
Multi-verse Optimizer
Supernova Optimizer
Simulated Annealing
Intelligent Water Drop
River Formation Dynamics
Wind Driven Optimization
Water Cycle Algorithm
Atmosphere Clouds Model Optimization
Artificial Raindrop Algorithm
Hurricane-based Optimization
Lightning Search Algorithm
Water Wave Optimization
Artificial Showering Algorithm
Circular Water Waves
Vortex Search Algorithm
Water Evaporation Optimization
Crystal Energy Optimization Algorithm
Lightning Attachment Procedure Optimization
Rainfall Optimization Algorithm
Hydrological Cycle Algorithm
Nuclear Fission–Nuclear Fusion
Nuclear Reaction Optimization
Whirlpool Algorithm
Electrostatic Discharge Algorithm
Targeted Showering Optimization
Harmony Search
Hysteretic Optimization
Electromagnetism-like Optimization
Particle Collision Algorithm
Central Force Optimization
Zombie Survival Optimization
Stem Cells Optimization Algorithm
Heart algorithm

Artificial Physics Optimization
Charged System Search
Chemical-reaction-inspired Optimization
Grenade Explosion Method
Ray Optimization
Mine Blast Optimization
Spiral Dynamics Algorithm
Gases Brownian Motion Optimization
Colliding Bodies Optimization
Kinetic Gas Molecule Optimization
States of Matter
General Relativity Search Algorithm
Optics Inspired Optimization
Stochastic Fractal Search
Heat Transfer Search
Ions Motion Algorithm
Sine Cosine Algorithm
Attraction Force Optimization
Electromagnetic Field Optimization
Ideal Gas Molecular Movement Algorithm
Thermal Exchange Optimization
Vibrating Particles System Algorithm
Sonar Inspired Optimization
Spring Search Algorithm
Flow Regime Algorithm
Coulombs Franklins Algorithm
Vapour-liquid Equilibrium-based Algorithm
Atom Search Optimization
Henry Gas Solubility Optimization
Artificial Electric Field Algorithm
Evolutionary Centers Algorithm
Ohm's Law Optimization algorithm
Electron Radar Search Algorithm
Spherical Search Optimizer
Equilibrium Optimizer
Newton Metaheuristic Algorithm
Tree Growth Algorithm
Farmland Fertility Algorithm
Tree Based Optimization
BladderWorts Suction Algorithm
Physarum-energy Optimization Algorithm
Fertile Field Algorithm
Clonal Selection Algorithm
Reincarnation Algorithm

Outline

1. Motivational Example
2. Representative Algorithms
    a. Newton Methods (Unconstrainted NLP)
    b. Evolutionary Algorithms (Unconstrained NLP)
    c. Lagrange Multipliers (Constrained NLP)
    d. IPOPT (Constrained NLP)
3. Examples in Energy Engineering
    a. Cascaded Dams Scheduling
    b. Wind Farm Layout Optimization

Theory of Lagrange Multipliers

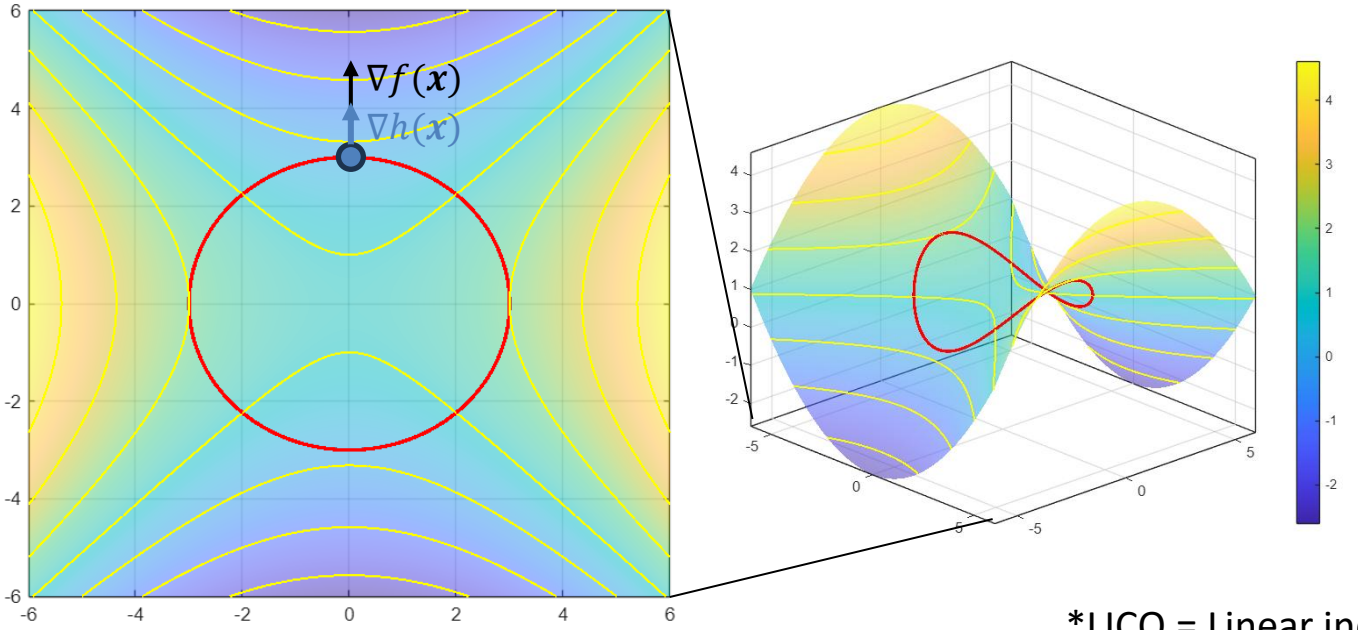
Consider a general **constrained NLP**:

Minimize: $f(\boldsymbol{x})$

Subject to: $h_i(\boldsymbol{x}) = 0 \quad i \in \mathcal{E}$

$g_j(\boldsymbol{x}) \leq 0 \quad j \in \mathcal{I}$

and $x_k^L \leq x_k \leq x_k^U \quad k = 1, 2, \ldots, n$



If $\boldsymbol{x}^*$ is a local minimizer and the *gradients of the active constraints* satisfy linear independence (LICQ), then there exists a set of scalars (Lagrange multipliers), λ_i $(i \in \mathcal{E})$ and σ_j $(j \in \mathcal{I})$ such that:

Karush-Kuhn-Tucker (KKT) Conditions

$$\nabla f(\boldsymbol{x}^*) + \sum_{i \in \mathcal{E}} \lambda_i \nabla h_i(\boldsymbol{x}^*) + \sum_{j \in \mathcal{I}} \sigma_j \nabla g_j(\boldsymbol{x}^*) = 0$$

or $\nabla_{\boldsymbol{x}} \mathcal{L}(\boldsymbol{x}^*, \boldsymbol{\lambda}^*, \boldsymbol{\sigma}^*) = 0$ *(stationarity)*

$h_i(\boldsymbol{x}^*) = 0$

$g_j(\boldsymbol{x}^*) \leq 0$ *(primal feasibility)*

$\lambda_i \geq 0, \quad i \in \mathcal{E}$

$\sigma_j \geq 0, \quad j \in \mathcal{I}$ *(dual feasibility)*

$\sigma_j g_j(\boldsymbol{x}^*) = 0, \quad j \in \mathcal{I}$ *(complementarity)*

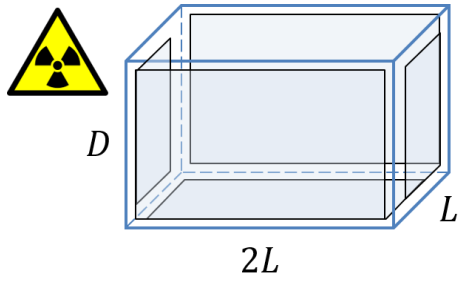
*LICQ = Linear independence constraint qualification

Theory of Lagrange Multipliers

Solve our previous example analytically using Lagrange Multipliers:

$$\min 1200(6DL + 2L^2)$$

$$2DL^2 = 8$$



Form the Lagrangian: $\mathcal{L}(D, L, \lambda) = f(D, L) - \lambda h(D, L)$

$$f(D, L) = 1200(6DL + 2L^2)$$

$$h(D, L) = 2DL^2 - 8$$

$$\mathcal{L}(D, L, \lambda) = 1200(6DL + 2L^2) - \lambda(2DL^2 - 8)$$

According to the KKT conditions:

$$\nabla f(\boldsymbol{x}^*) + \sum_{i \in \mathcal{E}} \lambda_i \nabla h_i(\boldsymbol{x}^*) + \sum_{j \in \mathcal{J}} \sigma_j \nabla g_j(\boldsymbol{x}^*) = 0$$

Solve for all gradients and equate to zero:

$$\frac{d\mathcal{L}}{dD} = 7200L - \lambda L^2 = 0$$

$$\frac{d\mathcal{L}}{dL} = 7200D + 4800L - 2\lambda DL = 0$$

$$\frac{d\mathcal{L}}{d\lambda} = -(DL^2 - 4) = 0$$

Solve for the system of 3 equations:

$$L = 1.82$$
$$D = 1.21$$
$$\lambda = 3962.31$$

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A Taxonomy of NLP Solvers

Zero-order Methods

Derivative-free, black-box

- Grid Search / Exhaustive Search
- Random Search
- Nelder-Mead Simplex

- Metaheuristic Search
  - **Genetic Algorithms**
  - **Particle Swarm**
  - Simulated Annealing
  - Differential Evolution
  - CMAES

- **Bayesian Optimization / Surrogate-based**

First-order Methods

Uses 1st derivative information

- Steepest Descent or Gradient Descent
- Stochastic Gradient Descent (SGD)
  - RMSProp, Adam, etc.
- Conjugate Gradient methods (e.g. Fletcher-Reeves)
- Coordinate Descent
- Generalized Reduced Gradient (GRG)

Second-order Methods

Uses 1st and 2nd derivative information

- *Newton's method*
- ***Quasi-newton method (BFGS)***

- Levenberg-Marquardt
- Trust Region Newton Methods
- Active Set Methods
- Sequential Quadratic Programming (SQP)
- **Interior Point Optimizer (IPOPT)**

Active Set vs. Interior Point Methods

Second-order Methods

Uses 1st and 2nd derivative information

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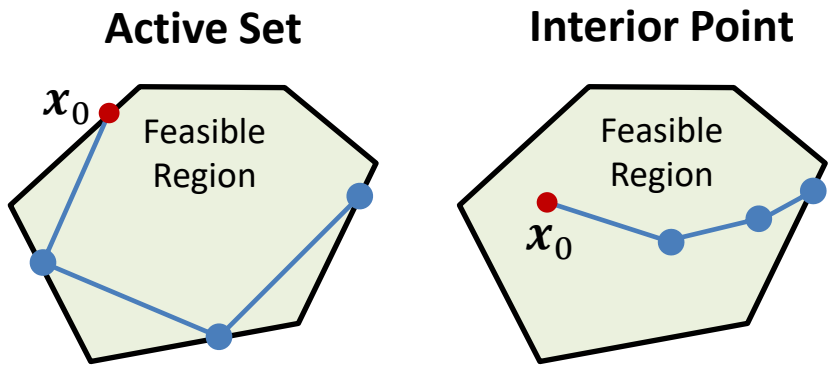
Active Set and Interior Point Methods

Minimize: $f(\boldsymbol{x})$

Subject to: $h_i(\boldsymbol{x}) = 0 \quad i = 1, 2, \ldots, l$

$g_j(\boldsymbol{x}) \leq 0 \quad j = 1, 2, \ldots, m$

and $x_k^L \leq x_k \leq x_k^U \quad k = 1, 2, \ldots, n$



Active Set Methods

- Popular since the 1970s
- **Idea:** Guess an initial active set of constraints, then add / delete using gradient and Lagrange multiplier information until optimality is detected.
- **Suitable for:** small number of constraints
- **Usage:** MATLAB SQP, Python scipy SLSQP, etc.

Interior Point Methods

- Popular since the 1990s
- **Idea:** Guess an initial $\boldsymbol{x}$ inside the feasible region, then stay strictly inside as you iterate until a boundary is approached.
- **Suitable for:** Large-scale, sparse problems
- **Usage:** IPOPT, MOSEK, other modern solvers

https://doi.org/10.1016/j.ejco.2025.100105

IPOPT (Interior Point OPTimizer)

https://github.com/coin-or/Ipopt

- Designed to find **local solutions** of optimization problems of the form:

$$\min_{\mathbf{x}\in\mathbb{R}^n} f(\boldsymbol{x})$$

$$\text{s.t.} \quad \boldsymbol{g}_L \le \boldsymbol{g}(\boldsymbol{x}) \le \boldsymbol{g}_U$$
$$\boldsymbol{x}_L \le \boldsymbol{x} \le \boldsymbol{x}_u$$

- The functions $\boldsymbol{f}(\boldsymbol{x})$ and $\boldsymbol{g}(\boldsymbol{x})$ can be nonlinear, nonconvex, but must be twice differentiable.

- Main steps:
  - Initialization
  - Check convergence
  - Compute search direction: **Damped Newton**
  - Solve for step size: **Filter Line Search**
  - Update

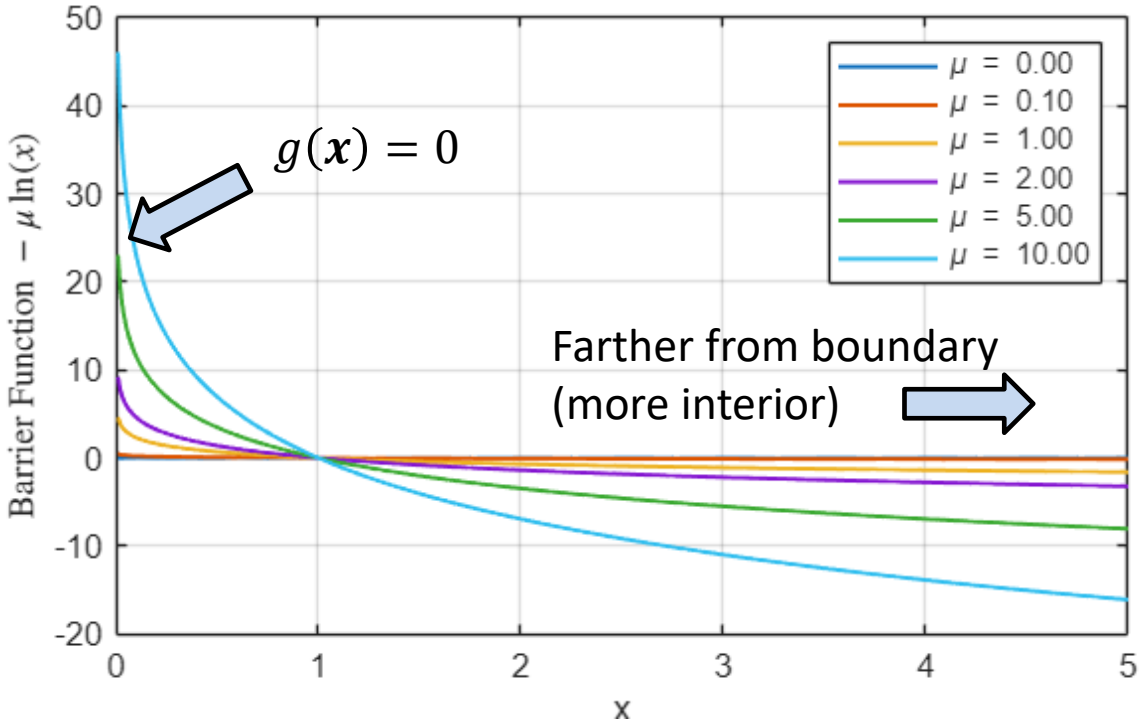
- A. Wächter and L. T. Biegler, On the Implementation of a Primal-Dual Interior Point Filter Line Search Algorithm for Large-Scale Nonlinear Programming, *Mathematical Programming* 106(1), pp. 25-57, 2006.

Barrier Method

Transform the NLP by adding a log-penalty, then decrease the value of *barrier parameter*, μ, to approach the original NLP.

$$\min_{\mathbf{x}\in\mathbb{R}^n} f(\boldsymbol{x}) \quad \Rightarrow \quad \min_{\mathbf{x}\in\mathbb{R}^n} f(\boldsymbol{x}) - \mu \sum_i \ln(g(\boldsymbol{x}_i))$$

$$\text{s.t.} \quad g(\boldsymbol{x}) \ge 0$$

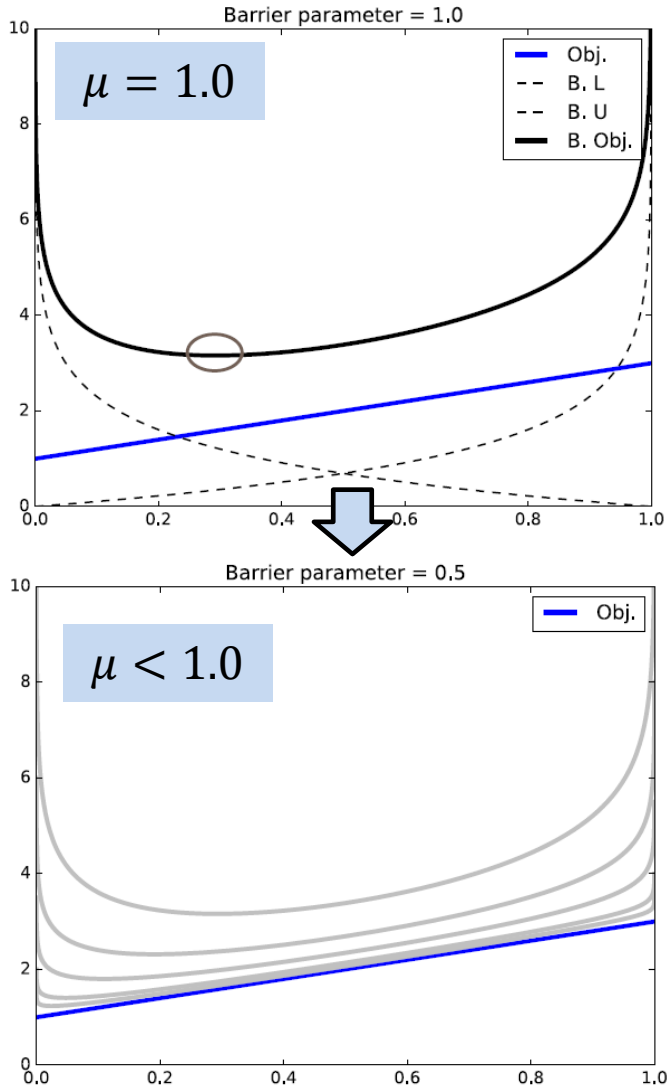


Source: https://www.youtube.com/watch?v=bJ0Kkf4u9bo

IPOPT (Interior Point OPTimizer)

https://github.com/coin-or/Ipopt

- Effect of the barrier term:

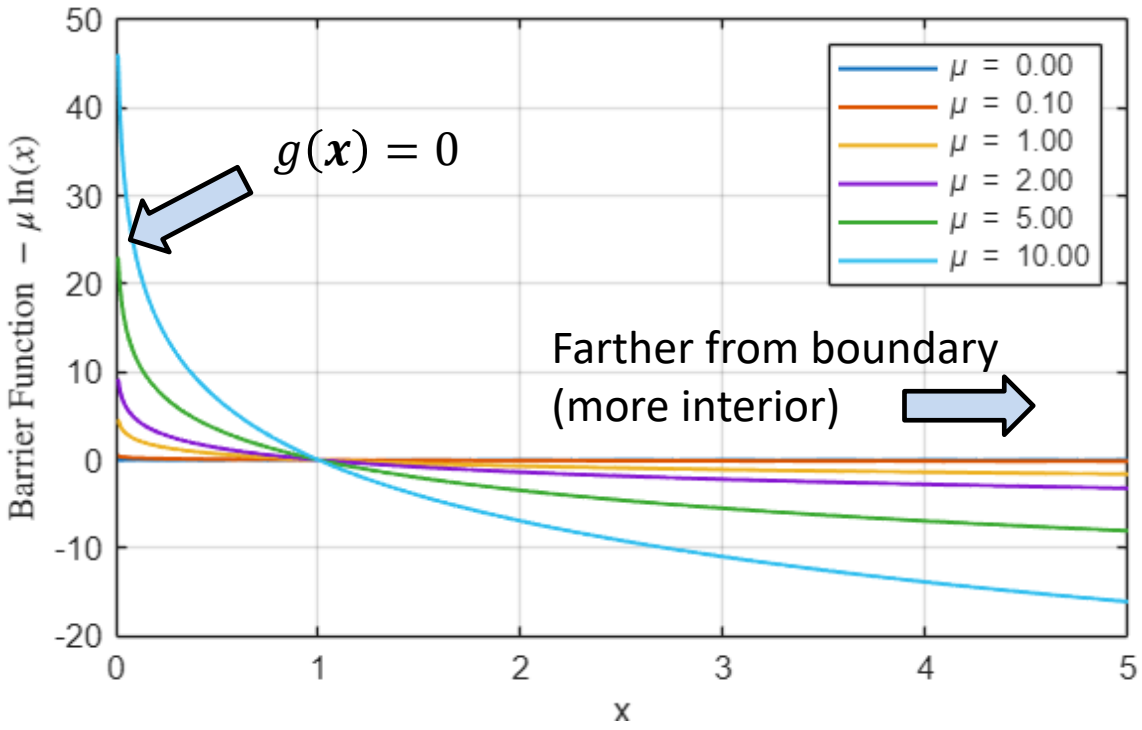


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s.t. $g(\boldsymbol{x}) \geq 0$



Source: https://www.youtube.com/watch?v=bJ0Kkf4u9bo

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